

T0-QAT: ξ -Aware Quantization-Aware Training

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Abstract

This document presents experimental validation of ξ -aware quantization-aware training, where $\xi = \frac{4}{3} \times 10^{-4}$ is derived from fundamental physical principles in the T0-Theory (Time-Mass Duality). The preliminary results show a relatively smaller performance drop under quantization noise than the standard approach, at a higher absolute error; the noise amplitude additionally contains a dataset-specific scaling factor.

.1 Introduction

Quantization-aware training (QAT) has emerged as a crucial technique for deploying neural networks on resource-constrained devices. However, current approaches often rely on empirical noise injection strategies without theoretical foundation. This work introduces ξ -aware QAT, grounded in the T0 Time-Mass Duality theory, which provides a fundamental physical constant ξ that naturally regularizes numerical precision limits.

.2 Theoretical Foundation

T0 Time-Mass Duality Theory

The parameter $\xi = \frac{4}{3} \times 10^{-4}$ is not an empirical optimization but derives from first principles in the T0 Theory of Time-Mass Duality. This fundamental constant represents the minimal noise floor inherent in physical systems and provides a natural regularization boundary for numerical precision limits.

The complete theoretical derivation is available in the T0 Theory GitHub Repository¹, including:

- Mathematical formulation of time-mass duality
- Derivation of fundamental constants
- Physical interpretation of ξ as quantum noise boundary

Implications for AI Quantization

In the context of neural network quantization, ξ represents the fundamental precision limit below which further bit-reduction provides diminishing returns due to physical noise constraints. By incorporating this physical constant during training, models learn to operate optimally within these natural precision boundaries.

.3 Experimental Setup

Methodology

We developed a comparative framework to evaluate ξ -aware training against standard quantization-aware approaches. The experimental design consists of:

- **Baseline:** Standard QAT with empirical noise injection
- **T0-QAT:** ξ -aware training with physics-informed noise
- **Evaluation:** Quantization robustness under simulated precision reduction

Dataset and Architecture

For initial validation, we employed a synthetic regression task with a simple neural architecture:

- **Dataset:** 1000 samples, 10 features, synthetic regression target
- **Architecture:** Single linear layer with bias
- **Training:** 300 epochs, Adam optimizer, MSE loss

.4 Results and Analysis

Quantitative Results

Interpretation

The experimental results demonstrate:

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Method	Full Precision	Quantized	Drop
Standard QAT	0.318700	3.254614	2.935914
T0-QAT (ξ -aware)	9.501066	10.936824	1.435758

Table 1: Performance comparison under quantization noise

- **Relatively Smaller Drop:** T0-QAT shows a 51 % smaller performance degradation from quantization (1.44 instead of 2.94)
- **Higher Absolute Error:** In both columns the error of T0-QAT lies above the standard approach (9.50 instead of 0.32 at full precision, 10.94 instead of 3.25 quantized, factor 3.4)
- **Noise Resilience:** Models trained with ξ -aware noise react less sensitively to precision variations in lower bits
- **Role of ξ :** The noise amplitude used is $\xi \cdot 80000 = 10.67$; the scaling factor 80000 is chosen per dataset, so ξ does not fix the amplitude by itself

.5 Implementation

Core Algorithm

The T0-QAT approach modifies standard training by injecting physics-informed noise during the forward pass:

```
# Fundamental constant from T0 Theory
xi = 4.0/3 * 1e-4

def forward_with_xi_noise(model, x):
    weight = model.fc.weight
    bias = model.fc.bias

    # Physics-informed noise injection
    noise_w = xi * xi_scaling * torch.randn_like(weight)
    noise_b = xi * xi_scaling * torch.randn_like(bias)

    noisy_w = weight + noise_w
    noisy_b = bias + noise_b

    return F.linear(x, noisy_w, noisy_b)
```

Complete Experimental Code

```
import torch
import torch.nn as nn
import torch.optim as optim
import torch.nn.functional as F

# xi from T0-Theory (Time-Mass Duality)
xi = 4.0/3 * 1e-4
```

```

class SimpleNet(nn.Module):
    def __init__(self):
        super().__init__()
        self.fc = nn.Linear(10, 1, bias=True)

    def forward(self, x, noisy_weight=None, noisy_bias=None):
        if noisy_weight is None:
            return self.fc(x)
        else:
            return F.linear(x, noisy_weight, noisy_bias)

# T0-QAT Training Loop
def train_t0_qat(model, x, y, epochs=300):
    optimizer = optim.Adam(model.parameters(), lr=0.005)
    xi_scaling = 80000.0 # Dataset-specific scaling

    for epoch in range(epochs):
        optimizer.zero_grad()
        weight = model.fc.weight
        bias = model.fc.bias

        # Physics-informed noise injection
        noise_w = xi * xi_scaling * torch.randn_like(weight)
        noise_b = xi * xi_scaling * torch.randn_like(bias)
        noisy_w = weight + noise_w
        noisy_b = bias + noise_b

        pred = model(x, noisy_w, noisy_b)
        loss = F.mse_loss(pred, y)
        loss.backward()
        optimizer.step()

    return model

```

.6 Discussion

Theoretical Implications

The relatively smaller drop of T0-QAT is a reason to examine whether physical principles can inform AI optimization strategies; with a higher absolute error and a freely chosen scaling factor it is not yet evidence. The ξ constant could provide:

- **Principled Regularization:** Physics-based alternative to empirical methods
- **Optimal Precision Boundaries:** Natural limits for quantization bit-widths
- **Cross-Domain Validation:** Connection between physical theories and AI efficiency

Practical Applications

- **Low-Precision Inference:** INT4/INT3/INT2 deployment (not yet tested)
- **Edge AI:** Resource-constrained model deployment

- **Quantum-Classical Interface:** Bridging quantum noise models with classical AI

Reproducibility

Complete code, experimental data, and theoretical derivations are available in the associated GitHub repositories:

- **Theoretical Foundation:**

Bibliography

- [1] Pascher, J. *T0 Time-Mass Duality Theory*. GitHub Repository, 2025.
- [2] Jacob, B. et al. *Quantization and Training of Neural Networks for Efficient Integer-Arithmetic-Only Inference*. CVPR, 2018.
- [3] Carleo, G. et al. *Machine learning and the physical sciences*. Reviews of Modern Physics, 2019.

0.1 Theoretical Derivations

Complete mathematical derivations of the ξ constant and T0 Time-Mass Duality theory are maintained in the dedicated repository. This includes:

- Fundamental equation derivations
- Constant calculations
- Physical interpretations
- Mathematical proofs